



Part I

Lab Assignment-3 (Introduction to Programming Structure)

Department: CSE
(22CPP209)

Subject: OOAD Lab

Objective: Introduction to templates *Array and Vector*, *Lvalue*, *Rvalue*, *references*, and *lambda function*.

Instruction: Explore the use of capture variable in your code when using lambda function.

Question 1: Write a C++ program that initializes a 10-element raw integer array. To print each array element, use `std::for_each` (include header *algorithm*) with a **lambda** function. Create a second lambda function that will replace out each element in the array with its square. Once the items have been squared, print the updated array to confirm.

Question 2: Write a C++ program that creates a `std::vector<int>` containing the first 10 positive integers. Define a function that accepts an `int&` parameter and modifies it by multiplying it by 10. Use the function to modify the elements of the vector through a lambda function that binds to the vector's elements. Print the modified vector.

Question 3: Write a C++ program that demonstrates the use of forwarding references.
